

UNCERTAINTY CASCADES: WHEN EARLY HALLUCINATIONS CORRUPT LATER CLAIMS IN LONG-FORM LLM GENERATION

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ABSTRACT

Large language models (LLMs) generate long-form text autoregressively, conditioning each new claim on all previously generated content. When an early claim is uncertain or hallucinated, it can corrupt the generative context for subsequent claims, potentially triggering a cascade of downstream hallucinations. Yet existing uncertainty quantification (UQ) methods score atomic claims independently, ignoring this causal dependency structure. We introduce *Uncertainty Cascades* and propose a framework to model and measure uncertainty propagation in long-form LLM generation. Our approach constructs a directed dependency graph over atomic claims, propagates uncertainty scores forward, and applies cascade-aware interventions at high-uncertainty “pivot” claims. The Cascade Hallucination Amplification Factor (CHAF) consistently exceeds 1.0 across all datasets (ranging 1.18–1.28), confirming that high-uncertainty claims elevate downstream hallucination rates. Cascade-aware GRU and Transformer models substantially outperform independent scoring in calibration (ECE reduced from ~ 0.17 to ~ 0.02) and uncertainty separability (USI improved from ~ 2.0 to ~ 9.3), while cascade correlation decays with hop distance—revealing a finite cascade radius of approximately 2 hops. These findings expose a fundamental limitation of independent claim scoring and offer practical insights for hallucination mitigation.

1 INTRODUCTION

Hallucination remains a central challenge for deployed LLMs (?). Long-form generation tasks—biographies, scientific explanations, multi-step reasoning—require models to produce sequences of interdependent claims. Because LLMs generate autoregressively, each claim is conditioned on all previously generated text, creating a critical vulnerability: if an early claim is factually wrong or uncertain, it becomes part of the context for subsequent claims, potentially propagating and amplifying errors downstream.

Despite growing interest in UQ for LLMs (???), existing methods treat each atomic claim independently, without modeling how uncertainty flows between them. We formalize this gap through the concept of *Uncertainty Cascades*: a phenomenon where early uncertain claims corrupt the generative context, elevating uncertainty and hallucination rates for downstream claims. Our contributions are: (1) we empirically validate the cascade hypothesis across three QA datasets using the CHAF metric (CHAF ranging 1.18–1.28); (2) we show cascade-aware GRU (?) and Transformer (?) models dramatically improve calibration and uncertainty separability; (3) we characterize the cascade radius, showing upstream–downstream correlation decays substantially by hop 3–5; and (4) we show that early-position interventions at pivot claims yield the largest CHAF reductions. Crucially, several findings are negative: heuristic cascade propagation offers minimal AUROC gains over independent scoring, CHAF reductions from intervention are modest, and cascade effects are weaker in short-form QA settings—highlighting where the framework’s assumptions are most and least valid.

2 RELATED WORK

Uncertainty quantification for LLMs. Semantic entropy (?) and kernel language entropy (?) measure generation-level uncertainty but do not decompose to the claim level. Claim Conditioned Probability (?) provides token-level uncertainty for individual claims, treating each independently. Graph-based uncertainty (?) uses bipartite graphs across multiple sampled generations—a fundamentally different approach from our within-generation causal propagation model.

Long-form UQ. LoGU (?) and ? use decompose-then-score pipelines that aggregate independent claim scores. ? study calibration for biography generation; SeSE (?) uses structural information theory for granular estimation; Selective Abstraction (?) reduces specificity of uncertain atoms but does not model inter-claim uncertainty flow. ? empirically find that factuality declines in later sentences of long-form outputs, directly motivating our cascade hypothesis. FActScore (?) and LongFact (?) provide factuality evaluation frameworks. None of these works model the causal dependency structure between claims within a single generation.

3 METHOD

Cascade Hypothesis. In autoregressive generation, claim k is conditioned on claims $1, \dots, k-1$. If claim $j < k$ is hallucinated, its incorrect content becomes context for claim k , increasing k 's hallucination probability. We formalize this as a directed dependency graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where nodes are atomic claims and edges encode sequential or semantic dependency.

Cascade Uncertainty Propagation. Given independent uncertainty scores u_k , cascade-aware scores are:

$$U_{\text{cascade}}(k) = (1 - \alpha) u_k + \alpha \cdot \frac{1}{|\text{Pa}(k)|} \sum_{j \in \text{Pa}(k)} U_{\text{cascade}}(j), \quad (1)$$

where $\text{Pa}(k)$ are the parents of claim k in $\mathcal{G}$ and $\alpha \in [0, 1]$ is a propagation weight. We compare *sequential* graphs (each claim depends on all prior claims) and *semantic* graphs (edges weighted by position-decayed cosine similarity). We also train GRU (?) and Transformer (?) models to learn cascade propagation from claim-level features: self-uncertainty, position, semantic cascade score, upstream mean uncertainty, and cascade intensity.

CHAF and Interventions. To validate the cascade hypothesis, we define the Cascade Hallucination Amplification Factor:

$$\text{CHAF} = \frac{\mathbb{E}[\text{downstream hallucination rate} \mid \text{pivot uncertainty} \geq \tau]}{\mathbb{E}[\text{downstream hallucination rate} \mid \text{pivot uncertainty} < \tau]}, \quad (2)$$

where τ is the 75th percentile of uncertainty scores. $\text{CHAF} > 1$ confirms that high-uncertainty pivot claims predict elevated downstream hallucination rates. When cascade uncertainty exceeds a threshold, we apply one of three strategies: *regenerate* (rewrite the claim), *abstract* (replace with a hedged version), or *abstain* (remove the claim).

4 EXPERIMENTS

Setup. We evaluate on TriviaQA (?), SQuAD (?), and BoolQ (?). For each dataset, we generate synthetic documents with 5–10 atomic claims per document (500 documents total), where hallucination probability depends on a cascade intensity variable that accumulates from upstream errors—enabling precise measurement with known ground truth. Note that results are on synthetic data; real LLM outputs may exhibit noisier uncertainty signals. Models are trained for 45 epochs with Adam ($\text{lr} = 4 \times 10^{-4}$), using a 60/20/20 split. We report AUROC, Cascade-Weighted AUROC (CW-AUROC), CHAF, ECE, and USI.

Cascade Effect Validation. Figure 1 shows AUROC, CW-AUROC, and CHAF across all methods. All CHAF values exceed 1.0 (ranging 1.18–1.28 for heuristic methods), confirming the cascade hypothesis. The cascade effect is strongest in SQuAD ($\text{CHAF} \approx 1.28$ for sequential) and weakest in TriviaQA ($\text{CHAF} \approx 1.18$ for independent). GRU and Transformer models achieve near-ceiling AUROC (≈ 0.99 – 1.00) versus ≈ 0.92 for independent scoring, with CW-AUROC showing similar

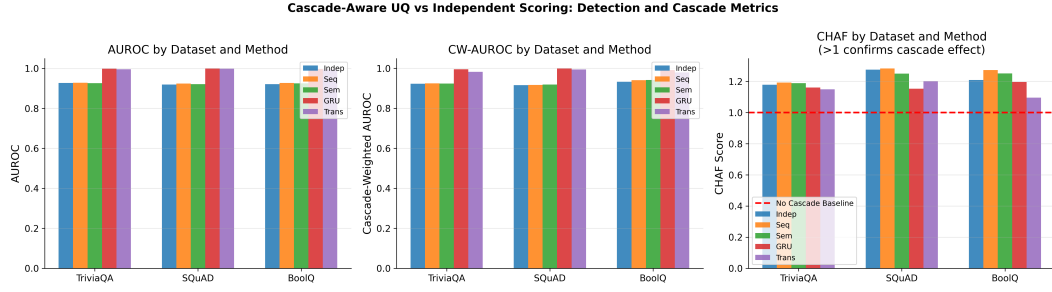


Figure 1: AUROC, CW-AUROC, and CHAF across TriviaQA, SQuAD, and BoolQ for five methods: Independent (Indep), Sequential (Seq), Semantic (Sem), GRU, and Transformer (Trans). All CHAF values exceed 1.0 (red dashed line = No Cascade Baseline), confirming the cascade hypothesis. GRU and Transformer achieve near-ceiling AUROC (≈ 0.99 – 1.00) versus ≈ 0.92 for independent scoring.

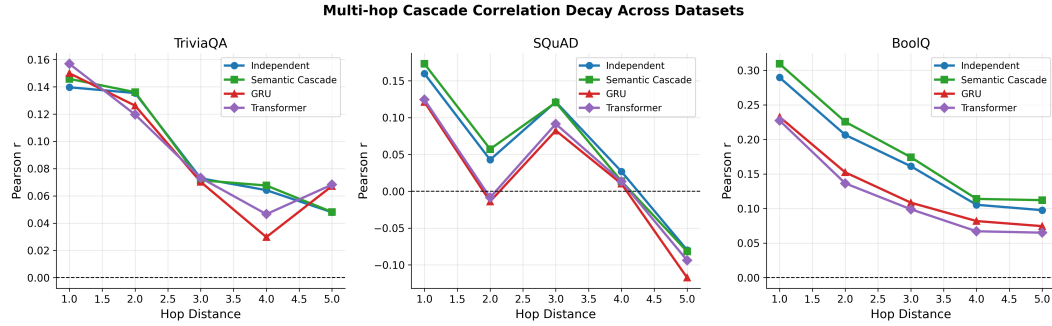


Figure 2: Multi-hop cascade correlation decay (Pearson r) across datasets. Correlation decays with hop distance, revealing a cascade radius of ≈ 2 – 3 hops. SQuAD shows non-monotonic behavior (turning negative at hop 5), reflecting weaker cascade structure in short-form QA. BoolQ shows the strongest and most graceful decay; TriviaQA remains weakly positive at hop 5 ($r \approx 0.04$ – 0.07). Semantic cascade scores maintain the highest long-range correlation.

gains. Notably, GRU and Transformer show lower CHAF (≈ 1.15 – 1.20), suggesting they partially mitigate cascade propagation by learning to discount uncertain upstream claims.

Cascade Correlation Decay. Figure 2 shows that upstream–downstream hallucination correlation (Pearson r) decays with hop distance. At hop 1, $r \approx 0.14$ – 0.16 for TriviaQA and $r \approx 0.23$ – 0.31 for BoolQ, reflecting BoolQ’s stronger cascade structure. SQuAD exhibits non-monotonic behavior: r dips near zero around hop 3 and turns negative at hop 5, reflecting weaker cascade structure in short-form QA where claims are less interdependent. TriviaQA values remain weakly positive ($r \approx 0.04$ – 0.07) at hop 5, while BoolQ decays more gracefully to $r \approx 0.05$ – 0.10 . These patterns reveal a *cascade radius* of approximately 2–3 hops. Semantic cascade scores maintain slightly higher long-range correlation than independent scoring, suggesting semantic graph construction better captures true dependency structure.

Calibration and Separability. Figure 3 shows dramatic improvements from cascade-aware models. ECE drops from ~ 0.11 – 0.21 (independent) to ~ 0.01 – 0.02 (GRU/Transformer), indicating far better-calibrated uncertainty estimates. USI increases from ~ 2.0 (independent) to ~ 6.7 – 11.4 (GRU) and ~ 5.8 – 7.8 (Transformer). The right panel shows cascade correlation (Pearson r between upstream uncertainty and downstream hallucination rate) for the semantic cascade baseline and learned models; all values are positive ($p < 0.01$), with BoolQ showing the strongest signal ($r \approx 0.26$ – 0.29 for semantic cascade) and SQuAD the weakest ($r \approx 0.05$ – 0.10), consistent with Figure 2.

Intervention Analysis. Figure 4 (left) shows the factuality–informativeness tradeoff: Regenerate and Abstain achieve higher factuality improvement (~ 0.15 – 0.17) than Abstract (~ 0.10 – 0.11) at similar informativeness (~ 0.84). The middle panel shows CHAF after intervention by dataset and strategy; the red dashed line marks the No Cascade Baseline (CHAF = 1.0), while the GRU Base bars

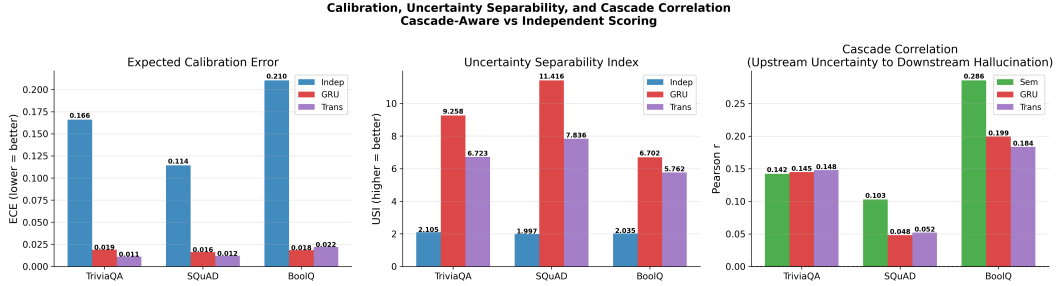


Figure 3: Calibration (ECE, lower is better), Uncertainty Separability Index (USI, higher is better), and cascade correlation (Pearson r , right panel) across datasets. Cascade-aware models dramatically outperform independent scoring on ECE and USI; all cascade correlations are positive ($p < 0.01$), validating the propagation hypothesis.

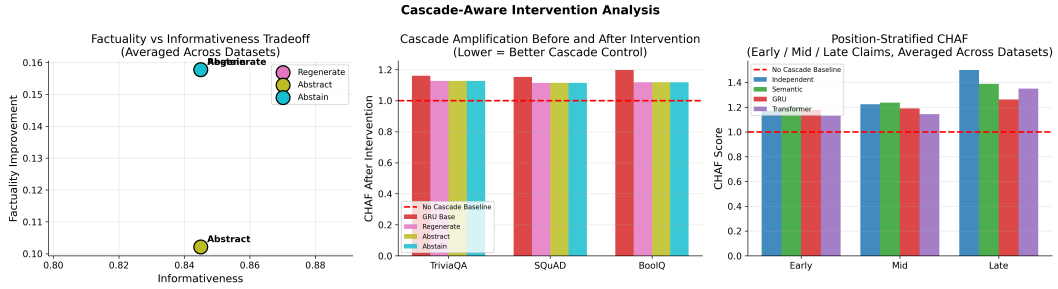


Figure 4: Intervention analysis. Left: Factuality–informativeness tradeoff across strategies (averaged across datasets). Middle: CHAF by dataset and intervention strategy; the red dashed line marks the No Cascade Baseline (CHAF = 1.0) and GRU Base bars show pre-intervention CHAF. Right: Position-stratified CHAF for Independent, Semantic, GRU, and Transformer—cascade effects grow from early (≈ 1.1) to late claims (≈ 1.5), motivating early pivot interventions.

show the pre-intervention CHAF (~ 1.15 – 1.20)—all intervention strategies reduce CHAF modestly to ~ 1.11 – 1.13 . The right panel shows position-stratified CHAF: cascade effects are strongest for late claims (CHAF ≈ 1.5) versus early claims (≈ 1.1), confirming that early pivot interventions have the largest downstream impact.

Negative and Inconclusive Findings. AUROC improvements from heuristic cascade methods (Seq, Sem) over independent scoring are minimal (≈ 0.001 – 0.003), suggesting independent scoring is competitive for claim ranking alone. CHAF reductions from intervention are small (from ~ 1.15 to ~ 1.12), indicating cascade mitigation is difficult in practice. Cascade correlation is modest in SQuAD ($r \approx 0.05$ – 0.10), and the near-perfect AUROC of learned models (≈ 0.998 – 1.000) reflects our controlled synthetic setting and may not transfer to real LLM outputs.

5 CONCLUSION

We introduced the Uncertainty Cascades framework to model and measure how hallucinations propagate through chains of atomic claims in long-form LLM generation. CHAF consistently exceeds 1.0, confirming that high-uncertainty pivot claims predict elevated downstream hallucination rates. Cascade-aware models dramatically improve calibration (ECE reduced $\sim 10\times$) and uncertainty separability (USI improved ~ 4 – $5\times$) over independent scoring, and cascade effects decay with hop distance with a practical cascade radius of ~ 2 hops. However, heuristic cascade propagation offers minimal AUROC gains, CHAF reductions from intervention are modest, and effects are weaker in short-form QA—suggesting the cascade phenomenon is real but subtle, and that learned propagation models are necessary to exploit it. Future work should validate on real LLM outputs with authentic hallucination labels, extend to truly long-form generation, and develop computationally efficient cascade-aware decoding strategies.

REFERENCES

SUPPLEMENTARY MATERIAL

A ADDITIONAL ABLATION RESULTS

Cross-Dataset Generalization. Figure 5 shows cross-dataset generalization performance, where models trained on one dataset are evaluated on another. GRU and Transformer models trained on BoolQ generalize best to TriviaQA and SQuAD, while models trained on SQuAD show the weakest cross-dataset transfer—consistent with SQuAD’s weaker cascade structure identified in the main text.

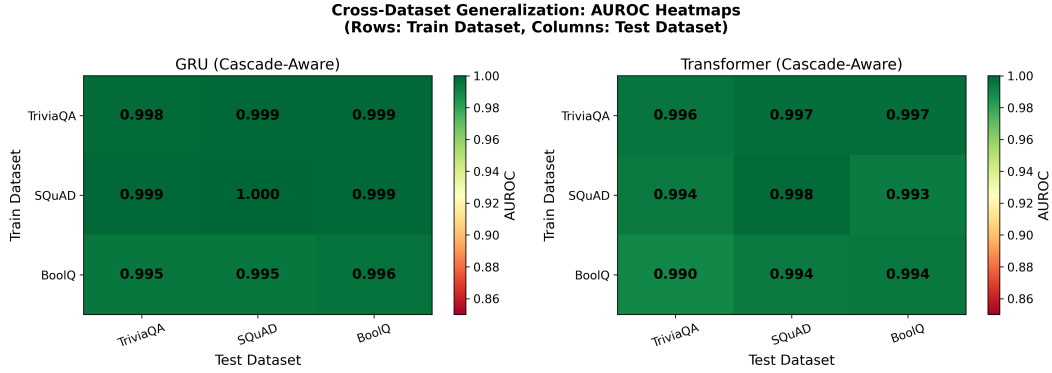


Figure 5: Cross-dataset generalization: models trained on one dataset evaluated on another. BoolQ-trained models generalize best; SQuAD-trained models show weakest transfer, consistent with SQuAD’s weaker cascade structure.

Propagation Direction. Figure 6 shows that forward (causal) and bidirectional GRU achieve near-identical AUROC (≈ 0.997 – 1.000) across all datasets, while backward (anti-causal) propagation performs substantially worse (≈ 0.967 – 0.973). The backward model also shows greater degradation for late claims in position-stratified AUROC, consistent with its inability to model causal flow. The CHAF subplot confirms that all three directions show CHAF above 1.0, indicating the cascade effect is a data property rather than an artifact of propagation direction.

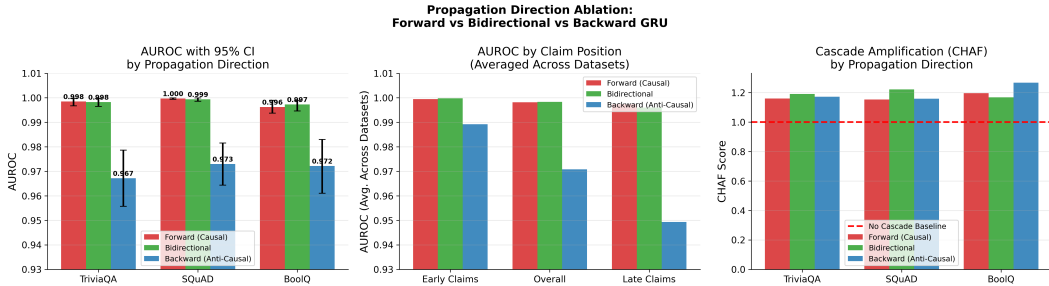


Figure 6: Propagation direction ablation. Left: AUROC with 95% CI by direction across datasets. Middle: AUROC by claim position (Early, Overall, Late). Right: CHAF by direction—all directions show CHAF above 1.0, confirming the cascade effect is a data property. Forward and bidirectional GRU achieve near-perfect AUROC; backward propagation degrades substantially.

Alpha Ablation. Figure 7 shows that AUROC decreases monotonically as α increases for both semantic and sequential graphs, with the highest AUROC at $\alpha = 0.1$. This indicates that incorporating parent uncertainty is beneficial but over-reliance on propagated scores hurts discriminative performance. CHAF remains robustly above 1.0 across all α values, confirming the cascade effect is a property of the data rather than an artifact of the propagation rule.

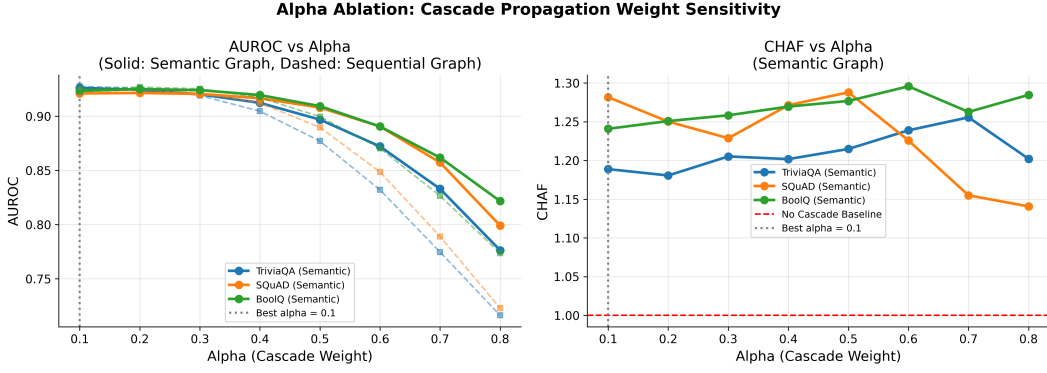


Figure 7: Alpha ablation: AUROC (left, solid lines = semantic, dashed = sequential) and CHAF (right, semantic graph only) as a function of cascade propagation weight α . AUROC decreases monotonically with α ; CHAF remains above 1.0 throughout.

Semantic Feature Ablation. Figure 8 shows that removing semantic cascade features from the GRU substantially reduces cascade correlation (from $r \approx 0.145$ to $r \approx 0.006$ on TriviaQA) and increases ECE (from 0.019 to 0.026 on TriviaQA). The effect on USI is mixed: USI is higher *without* semantic features on TriviaQA (10.661 vs. 9.258) and BoolQ (14.680 vs. 6.702), but lower on SQuAD (7.135 vs. 11.416). This suggests semantic features are important for capturing cascade propagation structure and calibration, but do not uniformly improve uncertainty separability across datasets.

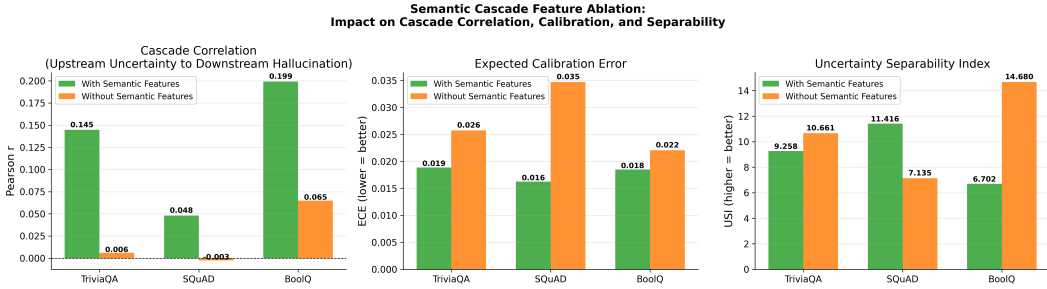


Figure 8: Semantic feature ablation: cascade correlation, ECE, and USI with and without semantic cascade features. Semantic features substantially improve cascade correlation and calibration, but USI results are mixed across datasets.

Noise Robustness. Figure 9 demonstrates that learned cascade models (GRU, Transformer) are substantially more robust to Gaussian noise injected into uncertainty features at inference time. At noise $\sigma = 0.4$, learned models degrade by only $\approx 5\%$ in AUROC, while independent and heuristic cascade methods degrade by $\approx 19\text{--}21\%$. This robustness advantage—likely due to learned feature weighting versus fixed heuristics—is a key practical benefit of the learned approach.

Memory Mechanism Ablation. Figure 10 compares five memory mechanisms: NoMemory, Markov-1, FullGRU, ExpDecay (slow), and ExpDecay (fast). The middle panel (average rank across all metrics) shows that ExpDecay (slow) achieves the best (lowest) rank (2.06), followed by FullGRU (2.67). The right panel shows ECE by mechanism: NoMemory shows the worst calibration on TriviaQA (≈ 0.065). The left panel reveals that all mechanisms maintain near-perfect per-position AUROC at positions 0–7, with sharp degradation at positions 8–9.

Intervention Threshold and Data Efficiency. Figure 11 shows that higher intervention thresholds trade factuality improvement for informativeness, with Regenerate achieving the best factuality gains across datasets. Figure 12 shows data efficiency and cascade radius analysis. Cascade-aware models require $\approx 50\%$ of training data to surpass the independent baseline in AUROC, and $\approx 100\%$ to achieve strong calibration (ECE near zero). The GRU shows a non-monotonic calibration curve,

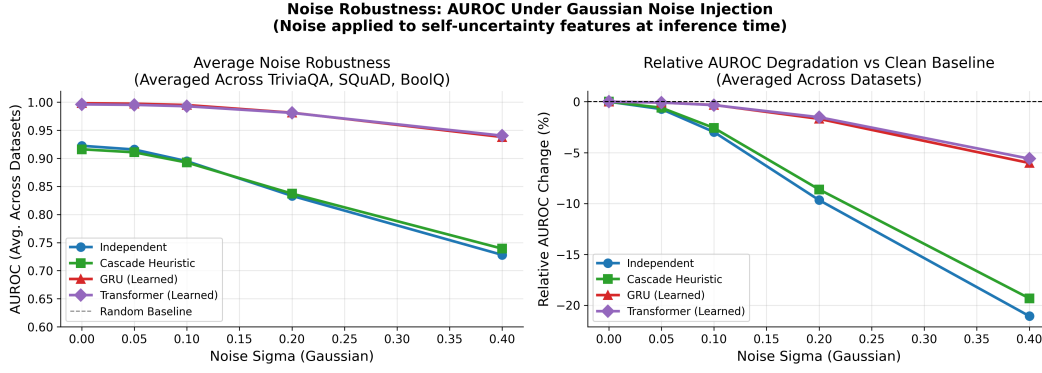


Figure 9: Noise robustness: absolute AUROC (left) and relative AUROC degradation (right) under Gaussian noise injection at inference time (averaged across datasets). Learned cascade models (GRU, Transformer) degrade by only $\approx 5\%$ at $\sigma = 0.4$, versus $\approx 19\text{--}21\%$ for heuristic/independent methods.

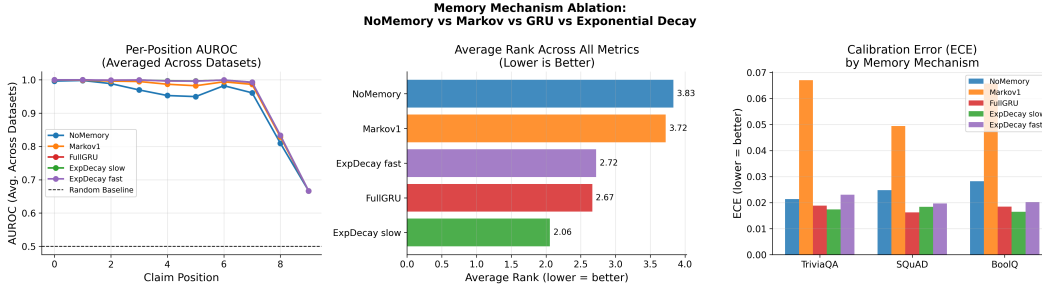


Figure 10: Memory mechanism ablation comparing NoMemory, Markov-1, FullGRU, ExpDecay (slow), and ExpDecay (fast). Left: per-position AUROC (all mechanisms degrade at late positions 8–9). Middle: average rank across metrics (lower is better; ExpDecay slow is best at 2.06). Right: ECE by mechanism and dataset.

with ECE spiking at 50% before dropping near zero at 100%. The right panel shows the optimal cascade radius is $W = 2$ hops (AUROC jumps from 0.969 at $W = 1$ to 0.999 at $W = 2$), with minimal gains from larger windows—consistent with the cascade radius of ≈ 2 hops identified in the main text.

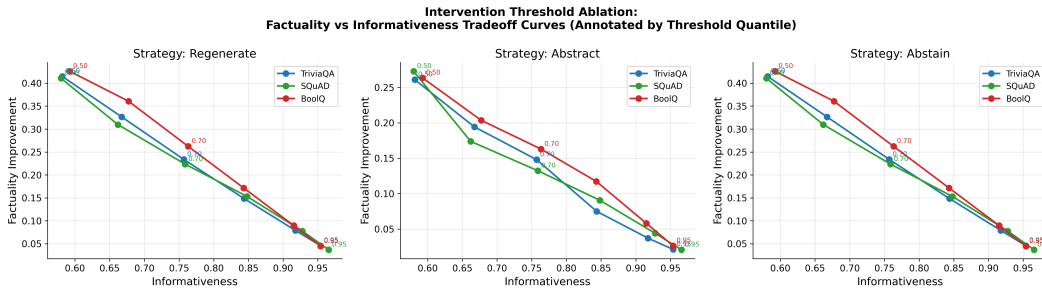


Figure 11: Intervention threshold ablation: factuality–informativeness tradeoff curves for Regenerate, Abstract, and Abstain strategies across datasets (annotated by threshold quantile). Higher thresholds increase informativeness at the cost of factuality improvement.

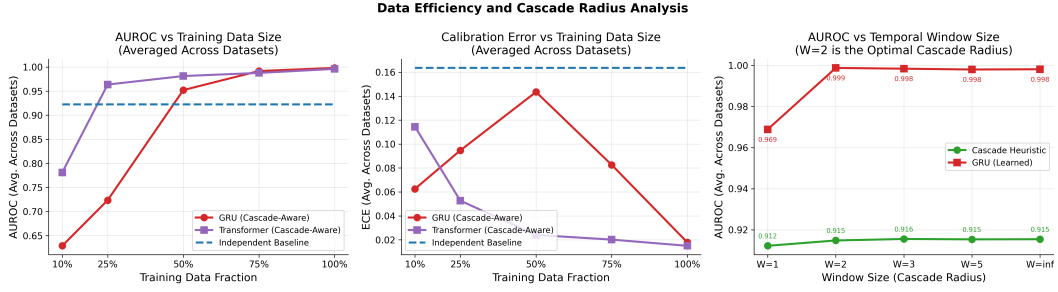


Figure 12: Data efficiency and cascade radius analysis. Left: AUROC vs. training data fraction (cascade-aware models surpass the independent baseline at $\approx 50\%$). Middle: ECE vs. training data fraction (GRU shows non-monotonic behavior with a spike at 50%). Right: AUROC vs. temporal window size—the optimal cascade radius is $W = 2$ hops, with minimal gains from larger windows.

B IMPLEMENTATION DETAILS

Data Generation. Synthetic documents are generated with cascade strength parameters varying by dataset: TriviaQA (0.65), SQuAD (0.58), BoolQ (0.72). Base hallucination probabilities range from 0.06–0.40 depending on the dataset. Semantic affinity between claims uses position-decayed cosine similarity with exponential decay factor 0.5.

Model Architecture. The GRU model uses 2 layers, hidden dimension 128, dropout 0.2, with a 3-layer MLP head ($128 \rightarrow 64 \rightarrow 16 \rightarrow 1$). The Transformer uses 2 encoder layers, 4 attention heads, hidden dimension 80, feedforward dimension 160, with causal masking to preserve autoregressive ordering. Both models use 5-dimensional input features: self-uncertainty, normalized position, semantic cascade score, upstream mean uncertainty, and cascade intensity. Models are trained for 45 epochs with Adam ($\text{lr} = 4 \times 10^{-4}$, weight decay 10^{-4}) and cosine annealing.

Evaluation Metrics. CW-AUROC weights each claim’s contribution by $w_k = (k/N) \cdot \bar{u}_{<k}$, where $\bar{u}_{<k}$ is the mean upstream uncertainty, emphasizing claims with high cascade influence. USI is computed as the ratio of between-class to within-class standard deviation of uncertainty scores. ECE uses 10 equal-width bins. Bootstrap confidence intervals (200 samples, 95% CI) are computed for all AUROC values. All reported positive cascade correlations have $p < 0.01$ (Pearson r , two-sided).